MARS

Malware Analysis & Reverse-engineering System

Analysis ID: MARS-20260619103407-C69918

Target: N/A

Generated: 2026-06-19 10:34:07

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1. File Intake Summary

Raw metadata captured by the intake module when the sample was submitted. Hash values are computed immediately on ingestion before any analysis modifies memory.

Analysis ID	`MARS-20260619103407-C69918`
Original File Name	`reverse_shell.exe`
File Size (Bytes)	`40850`
Extension	`.exe`
Creation Timestamp	`2026-06-19 07:49:03`
Magic Bytes (Hex)	`4D5A9000`
MIME/Format Guess	`application/x-msdownload (PE Executable)`
MD5	`0c411b04b00fb818b42ba31a0d5e1751`
SHA1	`8fc4a3868898c2b50d85e81b888d6bf8f5698cd8`
SHA256	`e7dea3de7e8ca278175fbea753cd56f8c76661909df630f9ee08c8c7b1a51878`
SHA512	`d9a7f9309e006f5e9cec572562d992d5a1812fd6b2df118505c511d9494a9889d491b8c398095bfa899febce4eb81cbe76e3fff91812eff667a6417b0af3270a`

2. Package & Archive Unpacking

Not applicable — the submitted file is not an archive or MSI package.

3. Deep Static Analysis

PE file structure analysis performed without executing the sample. Results include header metadata, security mitigations, section entropy, IAT inspection, manifest parsing, string extraction, and YARA signature matching.

Target: reverse_shell.exe

PE Headers

Machine Architecture	`0x14c` *Target CPU architecture encoded in the PE COFF File Header (e.g. 0x14c = x86).*
Compile Timestamp	`2026-04-04 17:54:16` *Date and time the binary was linked, taken from the PE File Header. Can be forged.*
DOS Header Offset (e_lfanew)	`0x80` *Byte offset of the PE signature from the start of the file (DOS stub field e_lfanew).*
Address of Entry Point	`0x12e0` *RVA of the first instruction executed; unusual values may indicate packing.*
Image Base	`0x400000` *Preferred virtual address at which the image is loaded into memory.*
Number of Sections	`13` *Count of PE sections; unusually high or low values can indicate packing.*

Mitigations

DEP / NX Bit	`Disabled` *Data Execution Prevention — prevents shellcode in data pages from executing.*

ASLR	Disabled *Address Space Layout Randomisation — randomises base addresses to hinder ROP chains.*
CFG (Control Flow Guard)	Disabled *Microsoft's control-flow integrity mechanism that validates indirect call targets.*
RFG (Return Flow Guard)	Disabled *Return-oriented-programming mitigation that validates return addresses.*
SafeSEH	Disabled *Structured Exception Handler protection — validates handlers against a compile-time table.*
Stack Canaries (/GS)	Disabled *Compiler-inserted stack cookie that detects stack-smashing overflows at runtime.*
Hardware Protection (CET/Shadow Stack)	Disabled *Intel CET Shadow Stack — hardware-enforced return-address integrity.*
Force Integrity	Disabled *Requires the binary to carry a valid Authenticode signature before loading.*
Isolation / AppContainer	Disabled *Binary runs inside an AppContainer sandbox with restricted privilege.*

Sections

Section: .text	Perms: R-E | Entropy: 6.19
Section: .data	Perms: RW- | Entropy: 0.22
Section: .rdata	Perms: R-- | Entropy: 4.79
Section: /4	Perms: R-- | Entropy: 4.73
Section: .bss	Perms: RW- | Entropy: 0.00
Section: .idata	Perms: RW- | Entropy: 4.57
Section: .CRT	Perms: RW- | Entropy: 0.11
Section: .tls	Perms: RW- | Entropy: 0.22
Section: /14	Perms: R-- | Entropy: 0.22
Section: /29	Perms: R-- | Entropy: 5.77
Section: /41	Perms: R-- | Entropy: 3.04
Section: /55	Perms: R-- | Entropy: 4.32
Section: /67	Perms: R-- | Entropy: 0.67

Suspicious Imports

Tracked APIs Found	0
APIs	None detected

Manifest Data

Manifest Status	Not Found
Requested Execution Level	Unknown

Strings Analytics

IPv4	1

IPv6	`0`
URL	`0`
Registry	`0` *Windows Registry key reads, writes, and deletions captured during execution.*
Email	`0`
Password-Like	`0`

Extracted Artifacts

IPv4	`['10.9.1.6']`
IPv6	`[]`
URL	`[]`
Registry	`[]` *Windows Registry key reads, writes, and deletions captured during execution.*
Email	`[]`
Password-Like	`[]`

YARA Signatures

Hits	`0`
Matched Rules	`Clean`

4. Dynamic Sandbox Analysis

Behavioural telemetry captured during live detonation inside the VMware sandbox. The agent instruments filesystem, registry, network, process, memory, and hardware subsystems in real time. Events are classified and counted per category.

No dynamic analysis results — either the dynamic module was skipped or the VMware sandbox is not configured.